SIMATSENGINEERING
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Engineer to Excel

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Course Code: MLA0112 **Slot: C**

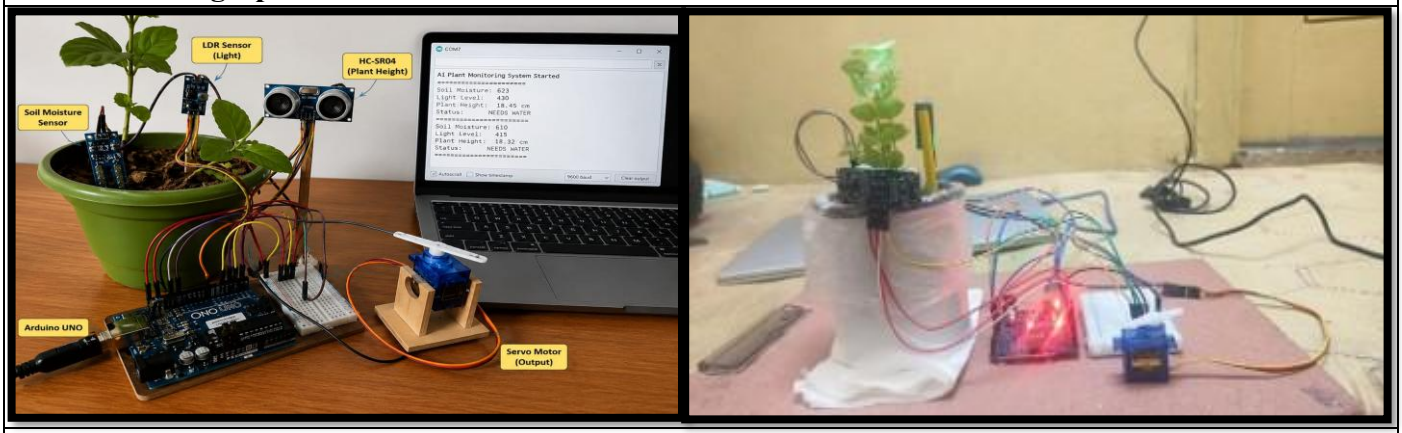
Course Name: Artificial Intelligence and Expert Systems for Image Transformation

Faculty: Dr. Logu &Dr. Prakash Mathialagan

Project Title:

AI-Based Smart Plant Monitoring and Automatic Irrigation System

Module Photographs:



Project Description:

The **AI-Based Smart Plant Monitoring and Automated Irrigation System** is an intelligent agricultural solution designed to monitor plant health and automate watering processes. The system uses multiple sensors, including a **soil moisture sensor**, **LDR sensor (light intensity)**, and **ultrasonic sensor (water level)**, connected to an **Arduino Uno** microcontroller. These sensors continuously collect real-time environmental data related to soil condition, sunlight availability, and water storage levels.

When the soil moisture level falls below a predefined threshold, the system automatically activates a **servo motor-controlled valve** connected to a water pump or water supply line. Water is delivered to the plant until the desired moisture level is reached, after which the valve is closed automatically to prevent overwatering. This automated process helps conserve water, reduces manual intervention, and promotes healthy plant growth.

The proposed system can be further enhanced by integrating **Artificial Intelligence (AI)** and **Internet of Things (IoT)** technologies for predictive plant health analysis, remote monitoring, and mobile notifications. The solution is suitable for home gardening, greenhouses, smart farming, and precision agriculture applications, offering an efficient, cost-effective, and sustainable approach to plant management.

Keywords: Artificial Intelligence, Smart Agriculture, Plant Monitoring, Automated Irrigation, Arduino Uno, Soil Moisture Sensor, LDR Sensor, Ultrasonic Sensor, Servo Motor, IoT.

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